1.7 RESISTORS IN SERIES-PARALLEL CIRCUIT

In a circuit, some resistors may be connected in series and some may be connected in parallel. In this, both series and parallel connection resistors are calculated separately and then the total resistance (Equivalent Resistance) can be calculated. Figure 1.26 shows R1& R2 are in parallel to each other and R3is series to R1 and R2.

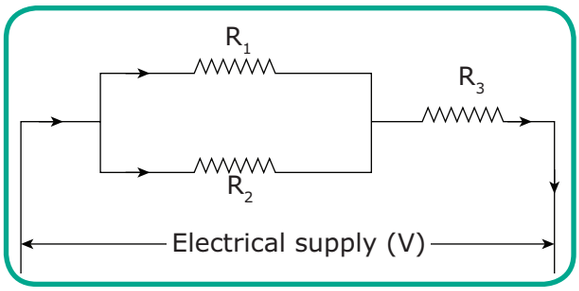


Figure 1.26 Resistors in series-parallel circuit

To find out the total parallel resistance Rp (the total resistance of parallel resistance only), the circuit is simplified as shown in Figure 1.27.

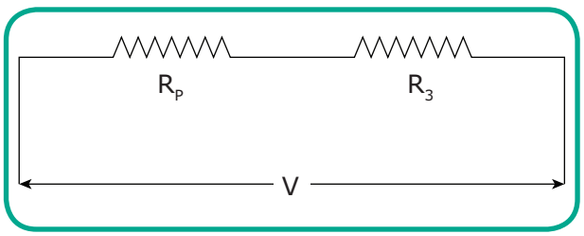


Figure 1.27

$$\frac{1}{R_P} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$R_P = \frac{R_1 R_2}{R_1 + R_2}$$

Now, R_p and R_3 are series as shown in Figure 1.27 for estimating the total resistance $R_T = R_p + R_3$.

Example: Find out the total resistance of the circuit given in Figure 1.28.

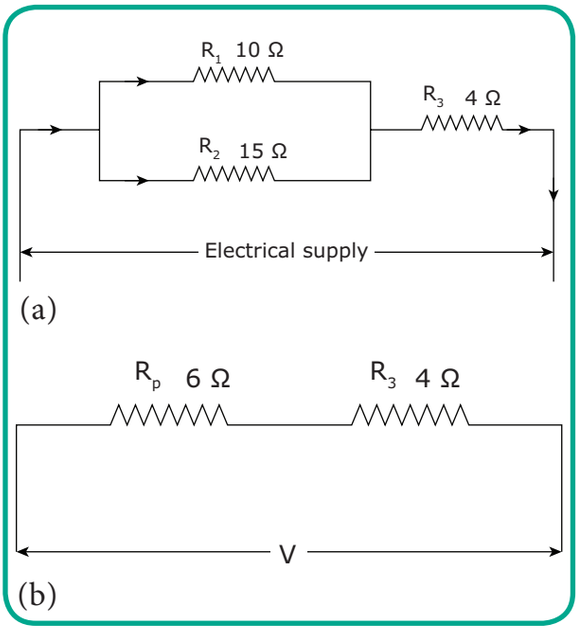


Figure 1.28

In the circuit shown in Figure 1.28,R_1 (10 Ω) andR_2 (15Ω) are parallel and R_3(4Ω) is series with the resistors R_1 and R_2.

$$R_P = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 15}{25} = \frac{150}{25} = 6\Omega$$

$$R_p = 6\ \Omega$$

$$R_T = R_p + R_3$$

$$R_T = 6+4$$

$$R_T = 10\ \Omega$$

Total resistance of the circuit is 10Ω

1.8 CLASSIFICATION OF RESISTORS

Resistors are broadly classified into two categories.

1. Fixed Resistor
2. Variable Resistor

1.8.1 Fixed Resistors

The value of fixed type of resistors cannot be changed. Hence, these resistors are called as fixed resistors as shown in Figure 1.29. Its symbol is shown in Figure 1.30.

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